

1/9/26
Day-18: DHCP-II

obtain IP
address automatically

50.0.0.0
50.1-50.100 (excluded)

50.10

Discover packet

(Send Discover
packet to DHCP server)

Transport
layer { SP: 68
DP: 67

N/w
layer { SIP: 0.0.0.0 [No IP assigned yet]
DIP: 255.255.255.255 [Flooded broadcast]

IP
Common IP for all
broadcast

D-L
{ Smac: client mac

Dmac: ffff → It cannot send ARP because
we don't know DIP.

offer

SIP: 50.0.0.10

Client IP is
not set

DIP: 255.255.255.255

request

SIP: 0.0.0.0

DIP: 255.255.255.255

(We know DIP of
DHCP here)

[We can't send ARP again directly
because it is unicast & we
should know both SIP & DIP.]

50.101-A
ACK

(Confirmation)

SIP: 50.0.0.10

DIP: 255.255.255.255

* After this only client
PC can receive IP.

So, when
DHCP server

- ① Discover
- ② Offer
- ③ Request
- ④ Ack

* If
process

DHCP

IP

com

N/w

HR

P

→

→

So, when a client wants to get IP from DHCP server, 4 packets are involved.

- ① Discover
 - ② Offer
 - ③ Req.
 - ④ Ack.
- } → DORA PROCESS

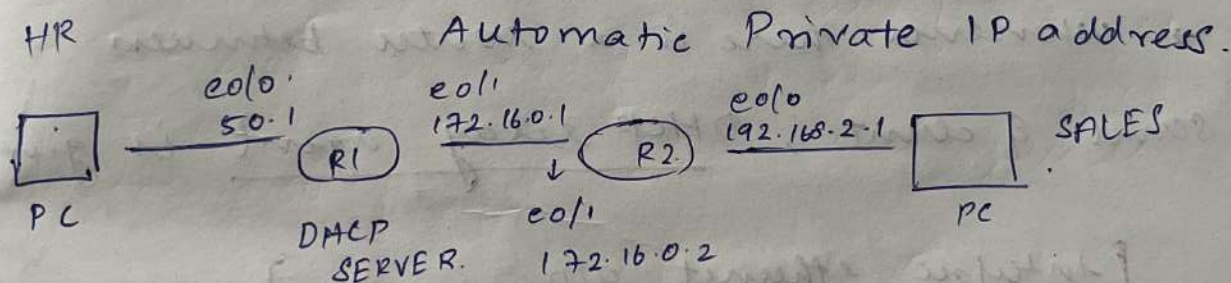
* If IP address is assigned then DORA process is done completely.

DHCP IMPLEMENTATION:

If DORA process fails, then windows can give IP directly from a particular N/w.

169.254.X.X → APIPA Network

↓



→ First we have to add static routes to R1 & R2

→ ip dhcp pool HR (any name)

network 50.0.0.0 255.0.0.0 sub net mask
↳ (64)

default-router 50.0.0.1 (gateway)

dns-server 50.0.0.30 [IP of DNS]

lease 1 2 3 [Days Hours Minutes] →

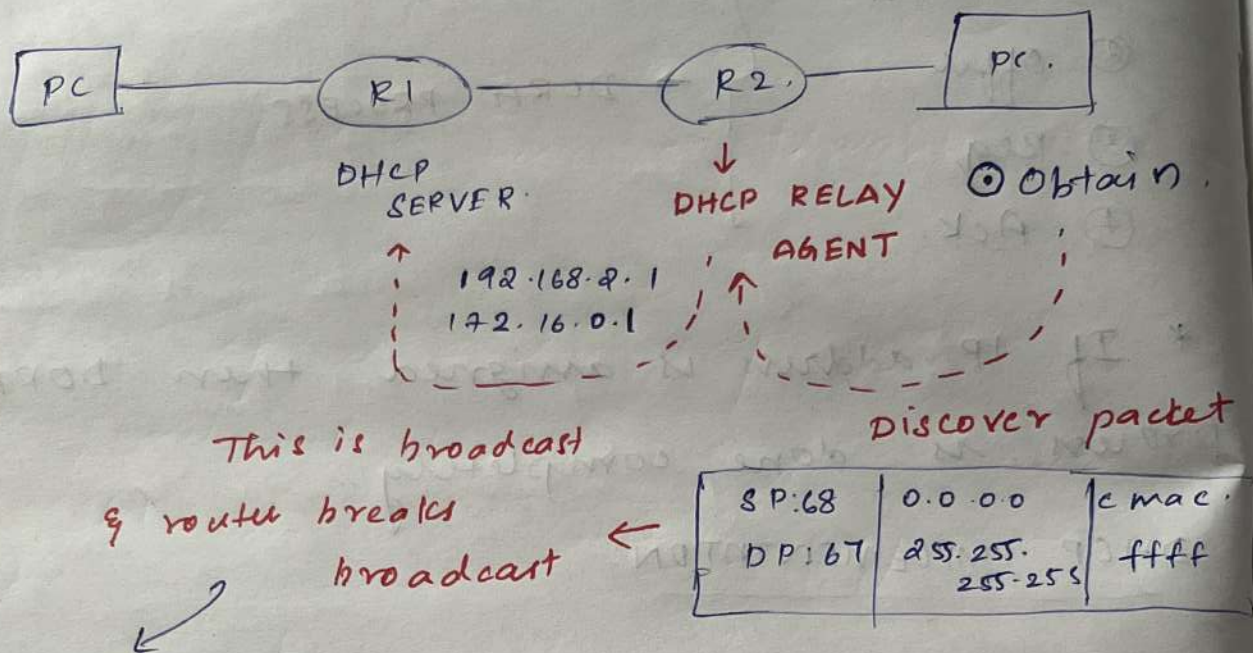
exit.

ip dhcp excluded-address 50.1 50.10

free IP
for static
routes

Command to see dhcp table:

show ip dhcp binding.



Because of this windows will give itself IP directly using APIPA network. 169.254.x.x

→ We should tell R2 to forward the packet

→ Whenever there is a router between server & client, DHCP relay agent is given

[interface ethernet 0/0
ip helper-address 172.16.0.1]

X * Whenever there is a router b/w client & server, DHCP relay agent will